

FIELD-TRIAL / THREE-COURSE MODEL

How the bed goes together.

A visual, step-by-step construction pack for the current 2400 × 1400 × 600 mm long-through model.

3

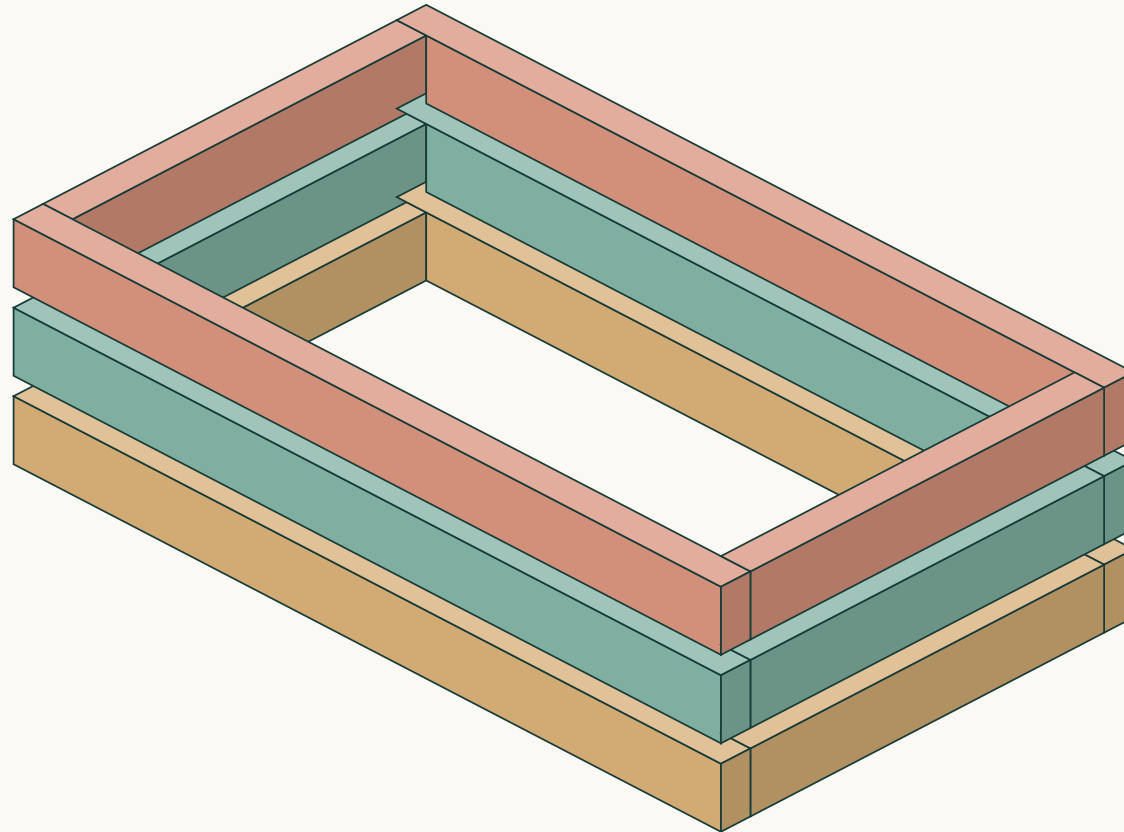
COURSES

12

TIMBER PARTS

56

FIXING ENTRIES

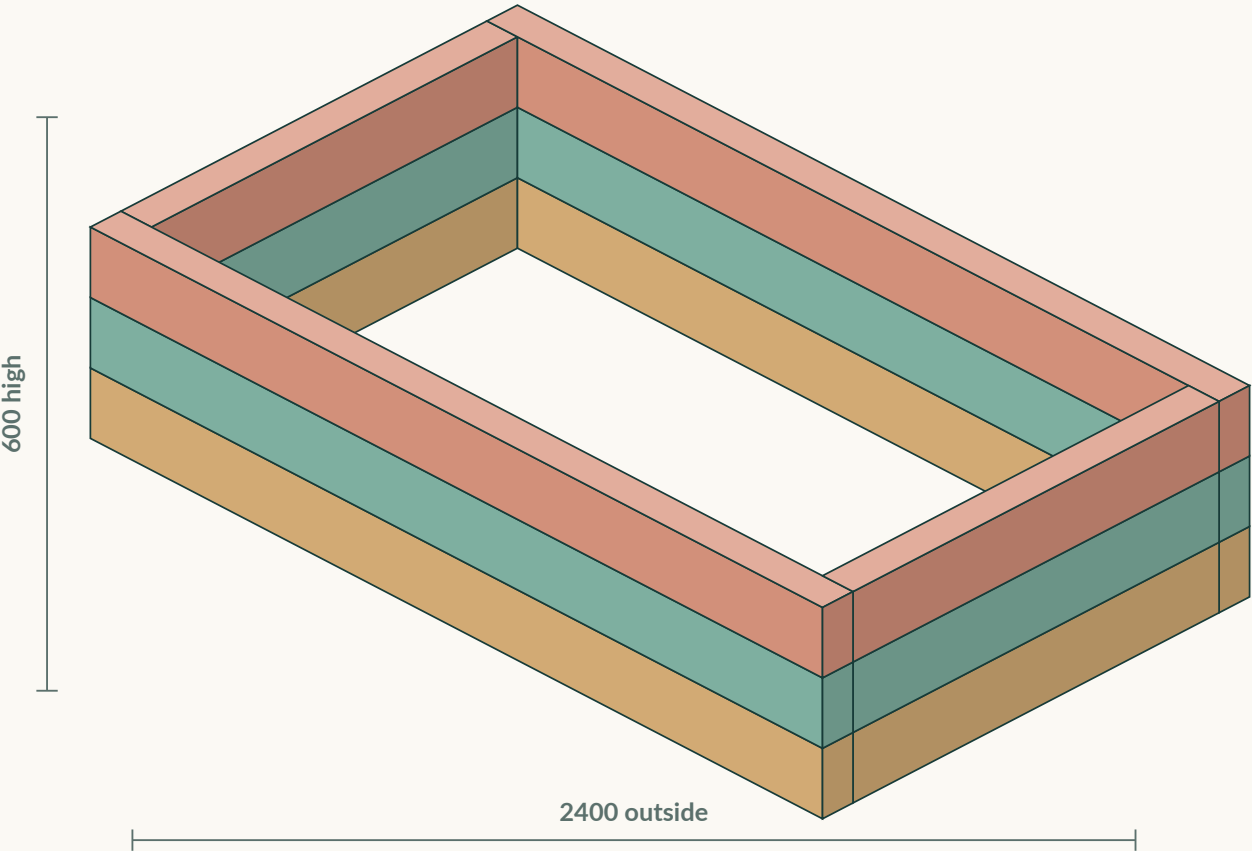
**IMPORTANT: CENTRES ARE KNOWN; DRILL SETTINGS ARE NOT.**

This pack shows the exact cut bands, part locations and screw entry centres in the current model. Pilot diameter/depth, screw-head seating and the site support/drainage detail are still unconfirmed. Use the visual sequence now; issue drilling only after those details are verified.

00 / WHAT YOU ARE BUILDING

One footprint, three identical rings.

Each course uses two 2400 long members and two 1200 cross-members. The long members pass through the corners on every course.



FINISHED GEOMETRY

ITEM	VALUE
Outside footprint	2400 × 1400
Total height	600
Clear opening	2200 × 1200
Course height	200
Wall thickness	100
Outside diagonal	2778.5
Geometric fill to 50 freeboard	1452 litres

CORNER ARRANGEMENT

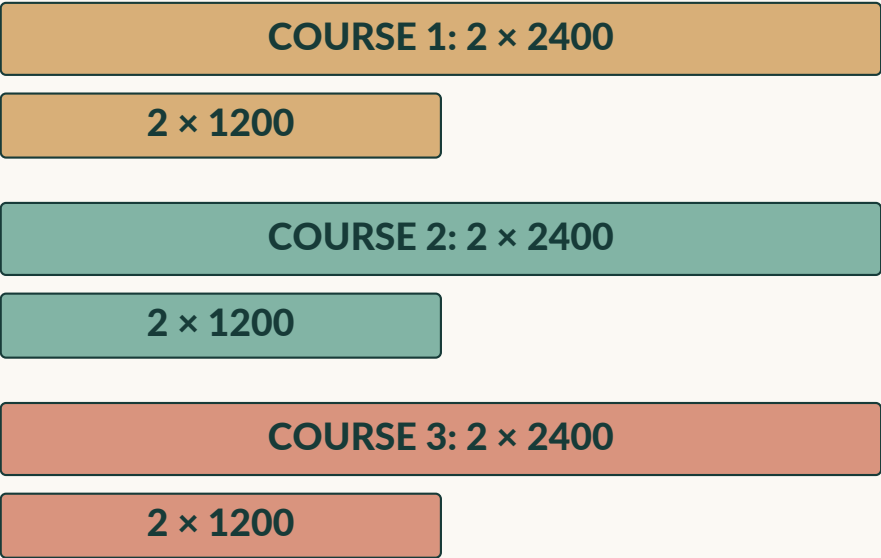
S and N are full-length through members on courses 1, 2 and 3. W and E fit between them. Do not switch to the older alternating-corner example.

01 / GET EVERYTHING ON THE FLOOR

Twelve pieces. Two screw families.

Lay the whole job out before cutting. Keep every stock-board ID tied to its resulting part.

TIMBER



FASTENERS



CORNER

24 fitted • 7 × 150 • Wickes 287686



STACK

32 fitted • 7 × 250 • Wickes 287688

SHOPPING QUANTITY

The model buys 6 × 2400 sleepers + 6 × 1800 sleepers. Screw allowance in the repo rounds to two packs of 25 of each screw family.

DO NOT MIX THE LABELS

The six 2400 boards are B001-B006. The six 1800 boards are B007-B012. Mark every board and its A end before you touch the saw.

02 / LABEL BEFORE CUTTING

A is the measuring origin.

Every cut and every fixing coordinate is measured from the labelled datum A of that stock or finished part.

A

→ U increases from A

Example: 2400 member

TOP face and **OUTER face** must remain identifiable after cutting and carrying.

STOCK DATUM ≠ PART DATUM

Cut coordinates are from the ORIGINAL stock A. Once a part is produced, its local A is the lower-X end for S/N or lower-Y end for W/E. Keep both meanings clear.

WHY THIS MATTERS

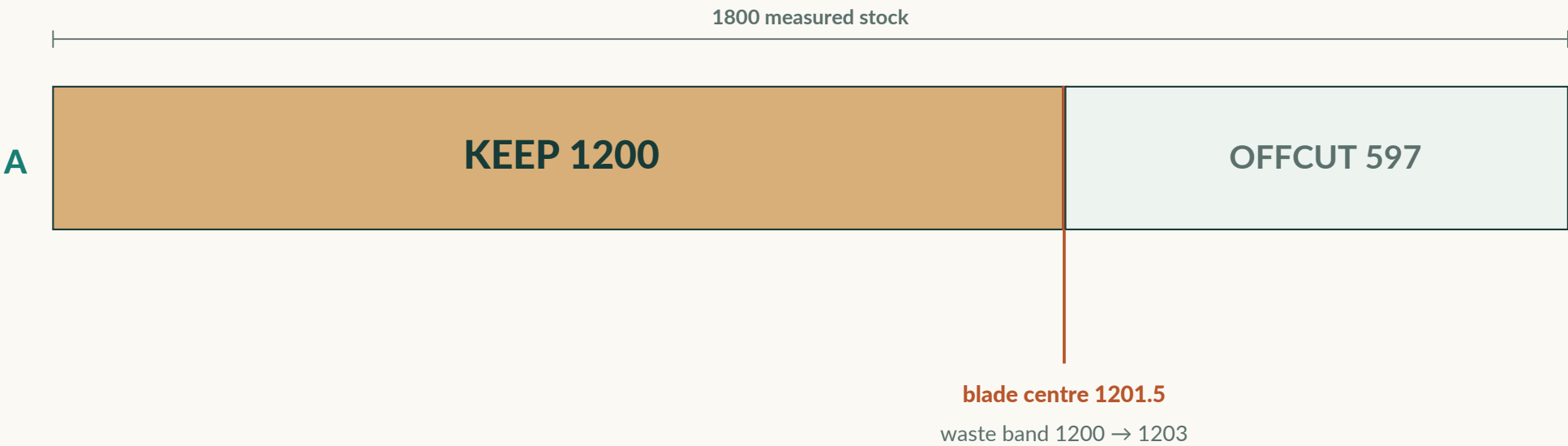
The course-3 vertical pattern is intentionally shifted. If a part is flipped end-for-end and measured from the wrong side, the stagger is destroyed even though every individual number still looks plausible.

Practical marking: write the board ID, A arrow, resulting part ID and TOP/OUTER arrows in pencil or removable crayon. Do not rely on “left/right as I am standing”.

03 / MAKE THE ONLY SIX CROSSCUTS

Keep 1200. Lose 3 mm of kerf.

The six long sides use sound 2400 stock uncut. Each 1200 cross-member comes from its own 1800 stock board.



BOARD	KEEP PART	CUT BAND
B007	C1-E / 1200	1200-1203
B008	C1-W / 1200	1200-1203
B009	C2-E / 1200	1200-1203
B010	C2-W / 1200	1200-1203
B011	C3-E / 1200	1200-1203
B012	C3-W / 1200	1200-1203

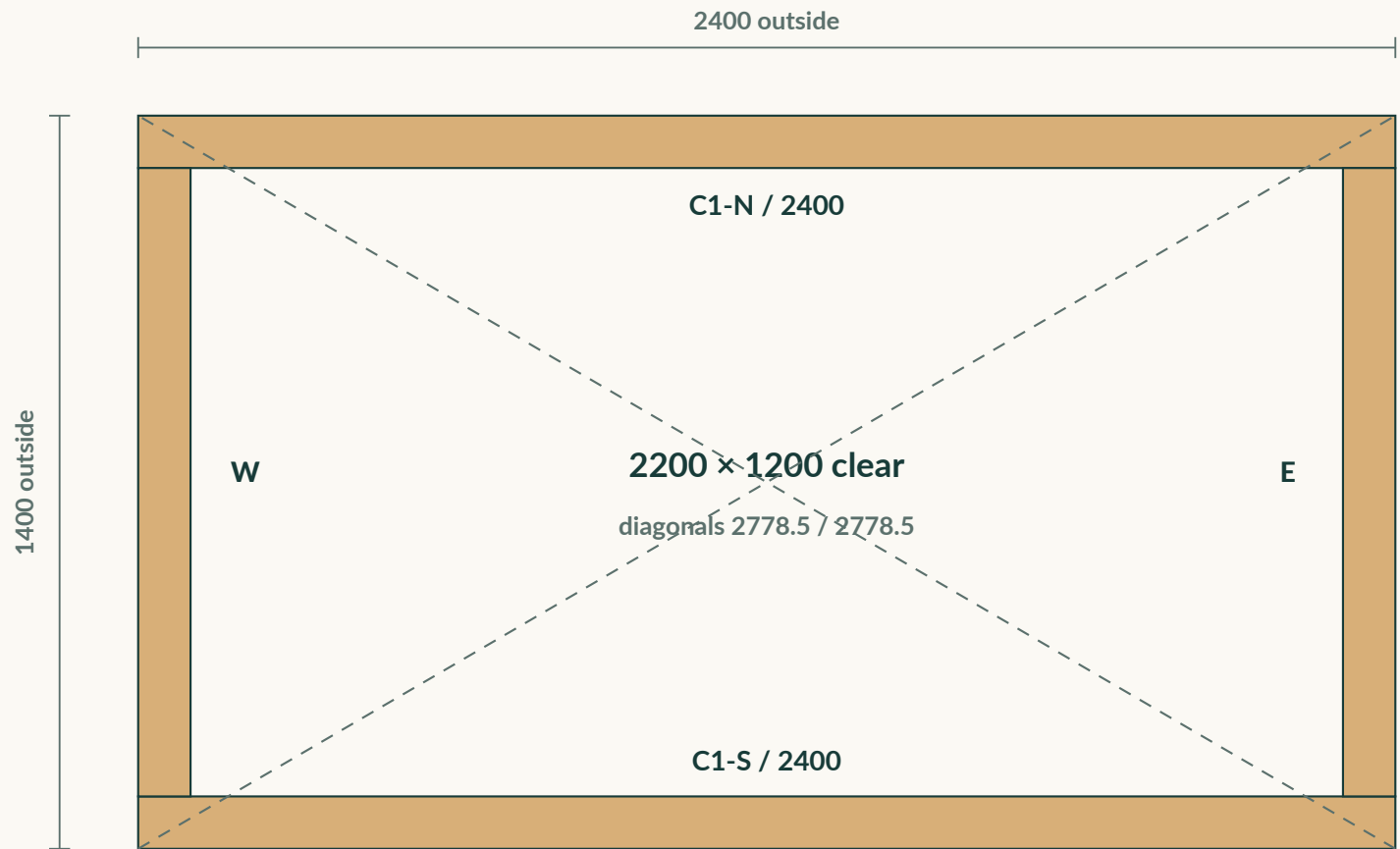
DO NOT CUT ONE 2400 INTO TWO 1200s

1200 + 3 + 1200 = 2403. The blade removes real material. The model deliberately uses separate 1800 stock for each exact cross-member.

04 / PLACE COURSE 1

Dry-fit the ring before fixing.

Lay S and N full length; fit W and E between them. Clamp the four joints. Square the outside rectangle.



CHECK BEFORE ANY HOLE

1) joints closed; 2) both diagonals equal; 3) long sides parallel; 4) base level. If the rectangle is wrong, do not let screws force it right.

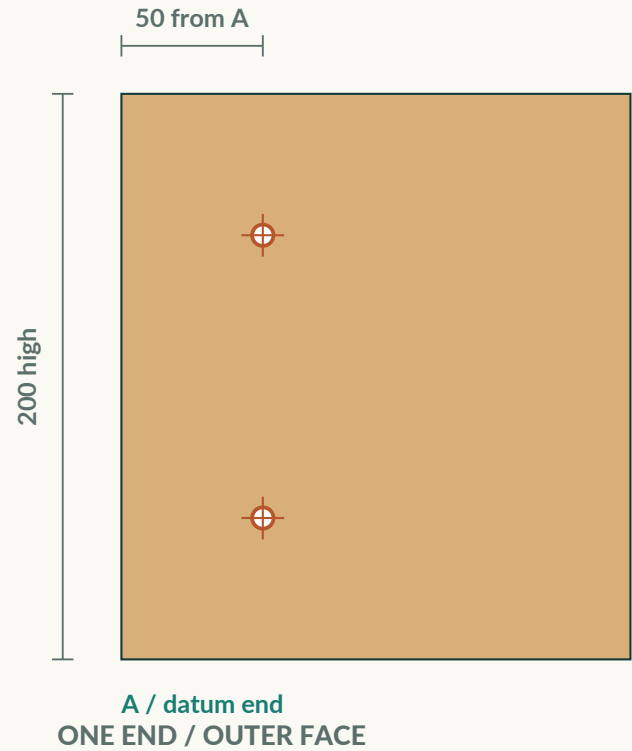
COURSE 1 HAS NO VERTICAL STACK SCREWS

Only the eight corner screws belong to course 1. Ground anchoring and supports are a separate design detail.

05 / MARK THE CORNER CENTRES

Four entries on each long member.

These centres repeat on S and N at every course. The short W/E pieces receive the screws in their end grain.



U FROM A	W UP	ENTRIES / LONG MEMBER
50 and 2350	50 and 150	4

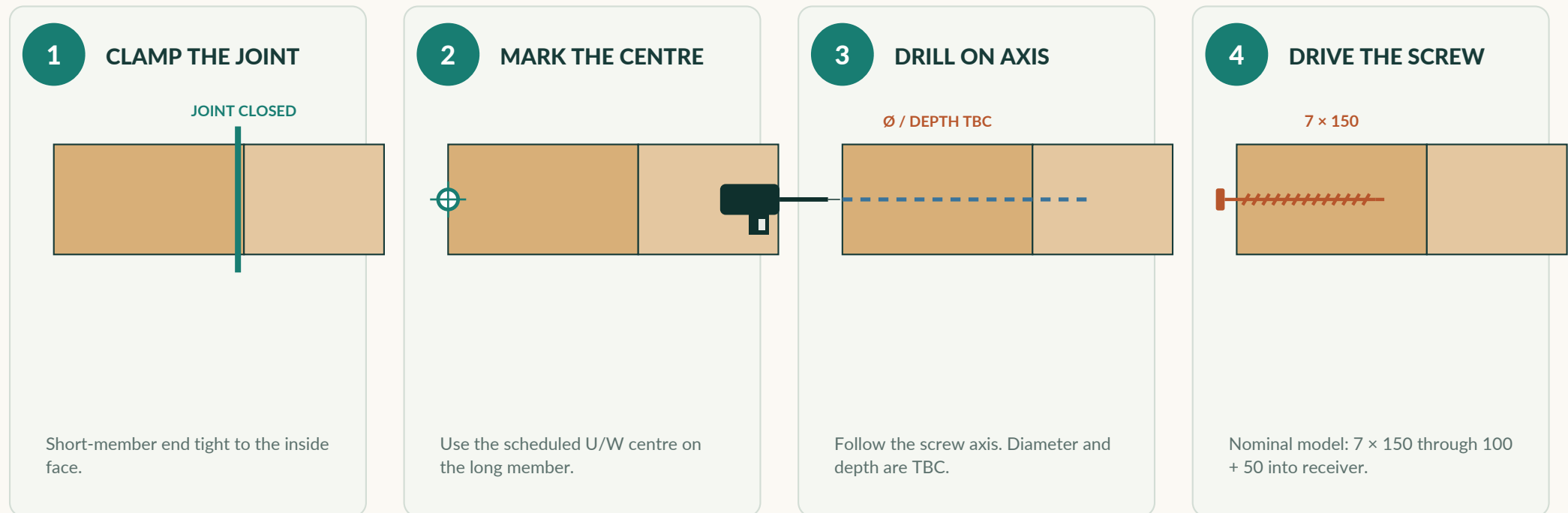
REPEAT THIS ON SIX LONG MEMBERS

C1-S, C1-N, C2-S, C2-N, C3-S and C3-N all use the same four corner centres. Total: 24 corner screws.

06 / CORNER: CLAMP → MARK → DRILL → DRIVE

One joint, four visible actions.

The drawing can show the complete sequence now. The exact pilot and head-seat settings remain a hard stop until verified.



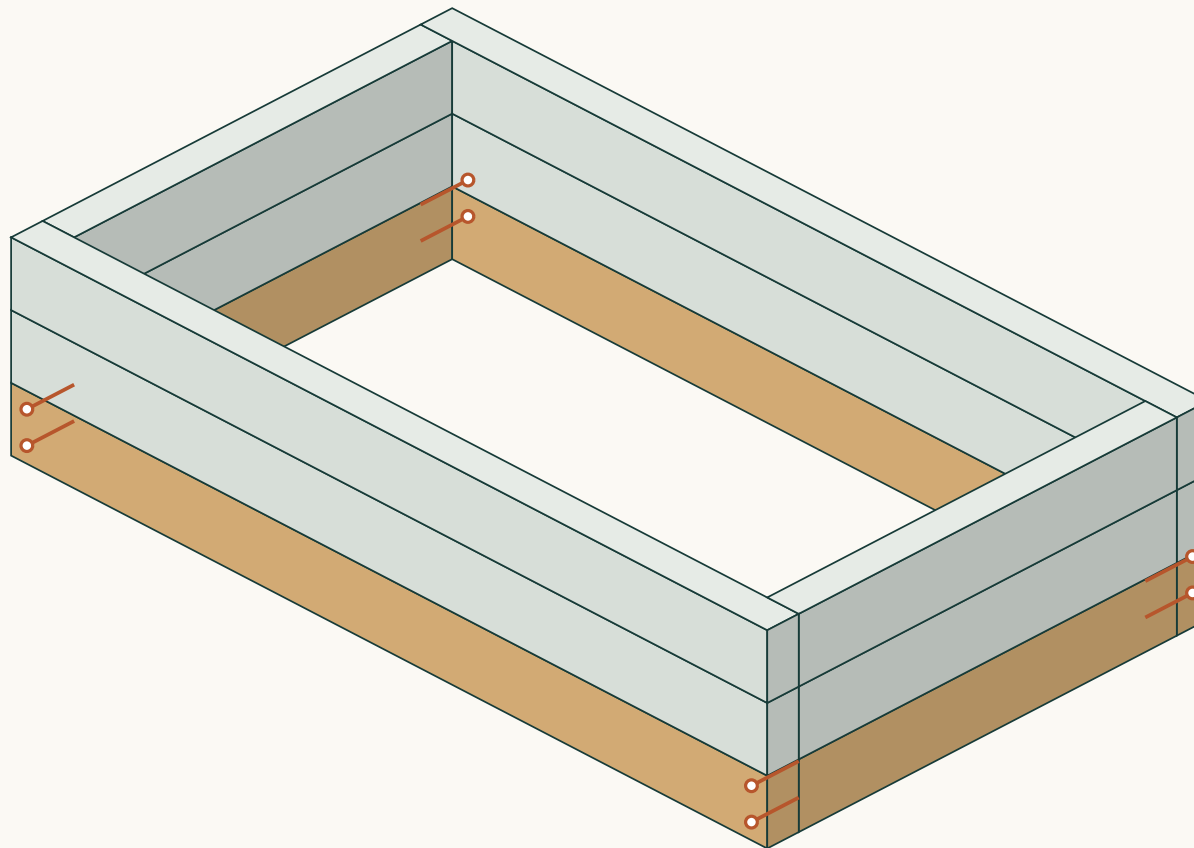
WHAT THE CROSS-SECTION MEANS

The 150 mm screw path is 100 mm through the long member + 50 mm nominal penetration into the receiving short-member end. That is screw-path geometry, not an issued pilot depth or proof of joint strength.

07 / FINISH COURSE 1

Eight corner screws complete the base ring.

Drive two screws at each corner only after the joint detail is approved and the frame is still clamped square.



8 × 7 × 150 CORNER SCREWS

AFTER DRIVING

Recheck 2400 × 1400, equal diagonals, level and joint closure. Record any movement before adding the next course.

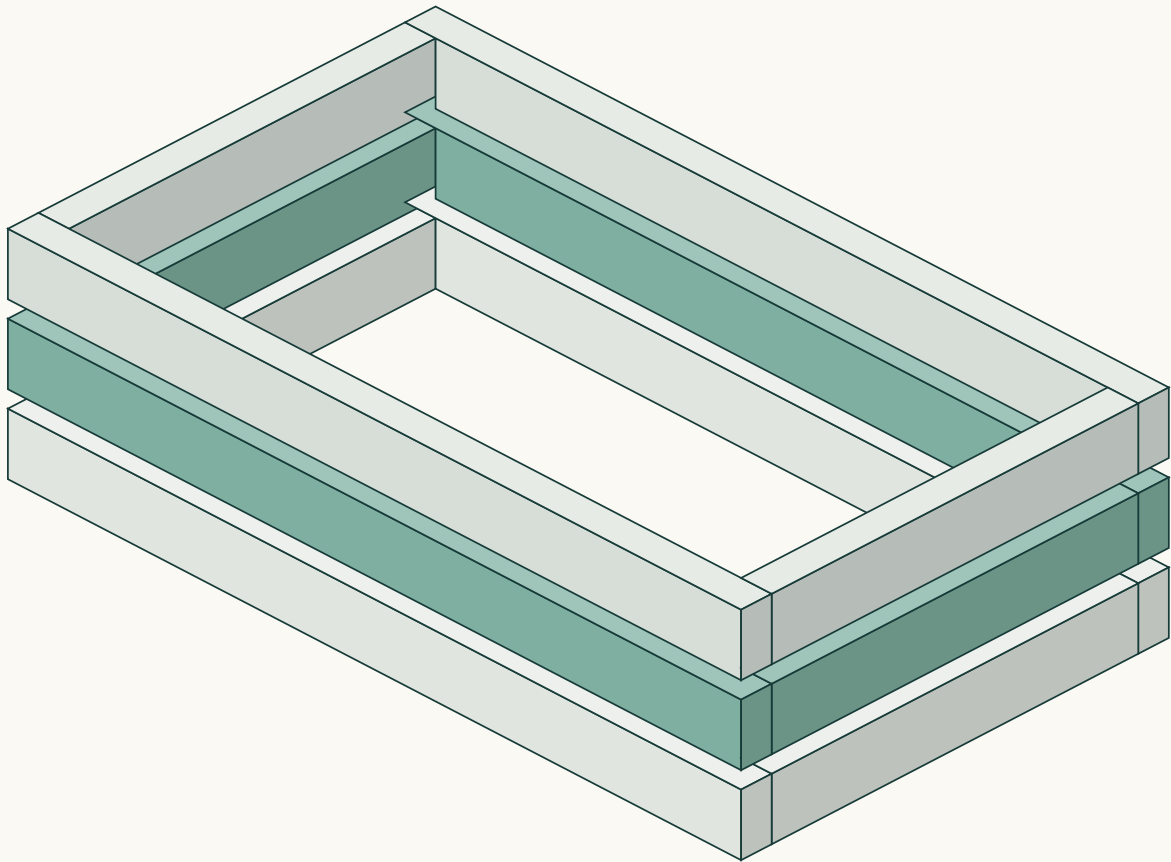
DO NOT HIDE A BAD JOINT

If the timber split, the head damaged the face or the joint opened, stop. Do not cover the evidence with course 2.

08 / PLACE COURSE 2

Put the next four members directly above course 1.

Same long-through corner pattern: S/N are 2400; W/E are 1200. Clamp the layer before marking the top fixings.



COURSE 2 PARTS

PART	LENGTH	BOARD
C2-S	2400	B004
C2-N	2400	B003
C2-W	1200	B010
C2-E	1200	B009

SEAT THE COURSE FULLY

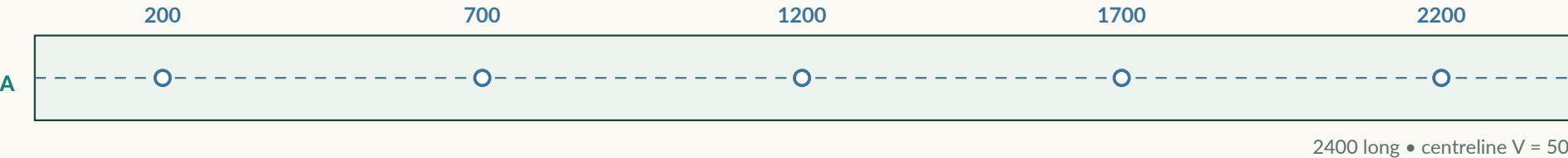
The upper course must bear on the timber below, not on debris or projecting hardware. Head seating is still an unresolved detail for any surface that another course will cover.

09 / MARK COURSE 2 TOP FIXINGS

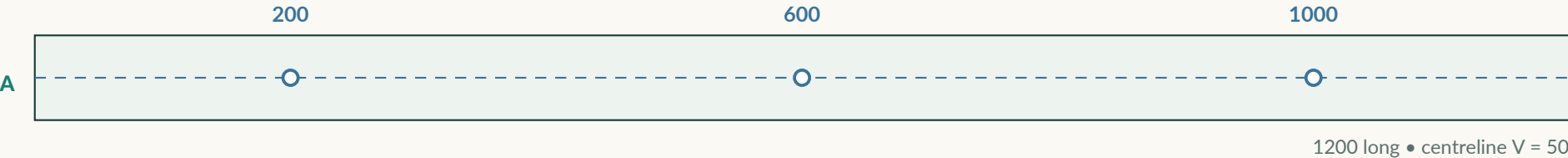
Sixteen vertical entries.

Every top entry is centred across the 100 mm face at V = 50. U is measured from each piece’s A end.

S + N / EACH 2400 MEMBER



W + E / EACH 1200 MEMBER



COURSE 2 TOTAL

5 + 5 + 3 + 3 = 16 stack screws. These are in addition to the eight ordinary course-2 corner screws.

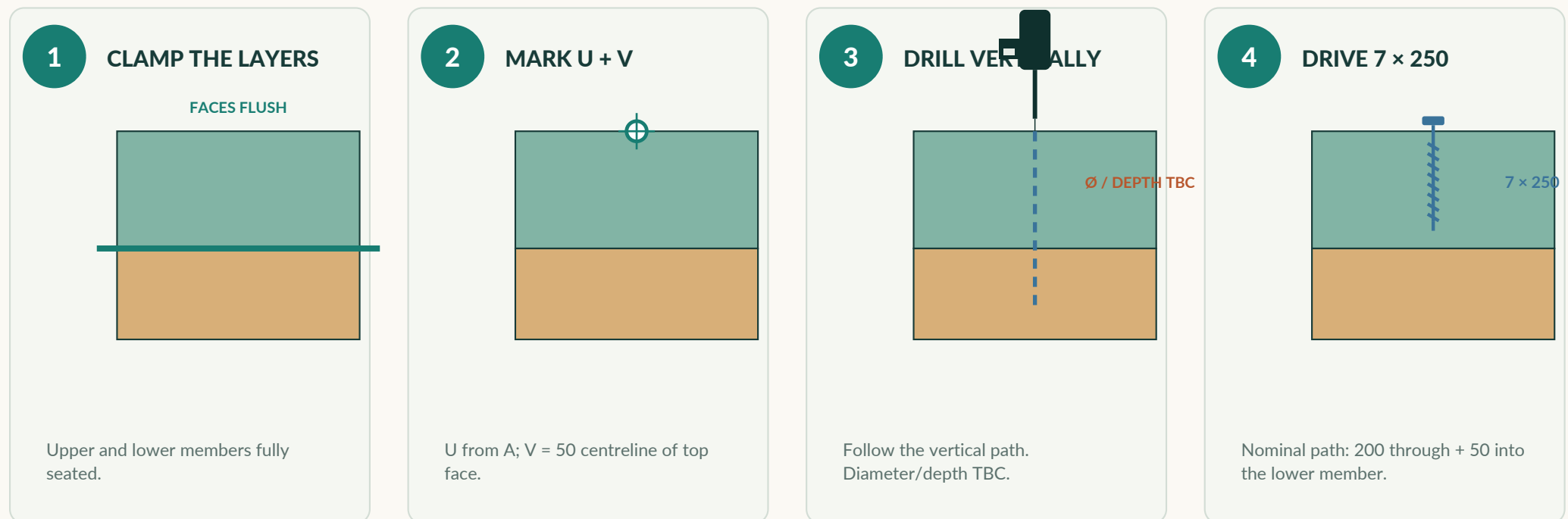
ENTRY HEIGHT

W = 200 on the upper piece: the screw starts on its TOP face and travels vertically down. The nominal 250 mm shaft crosses 200 mm of upper timber + 50 mm into course 1.

10 / STACK: CLAMP → MARK → DRILL → DRIVE

The vertical screw gets its own operation sequence.

The PDF shows the bit and screw moving down the same scheduled axis. The final head-seat condition still needs approval.



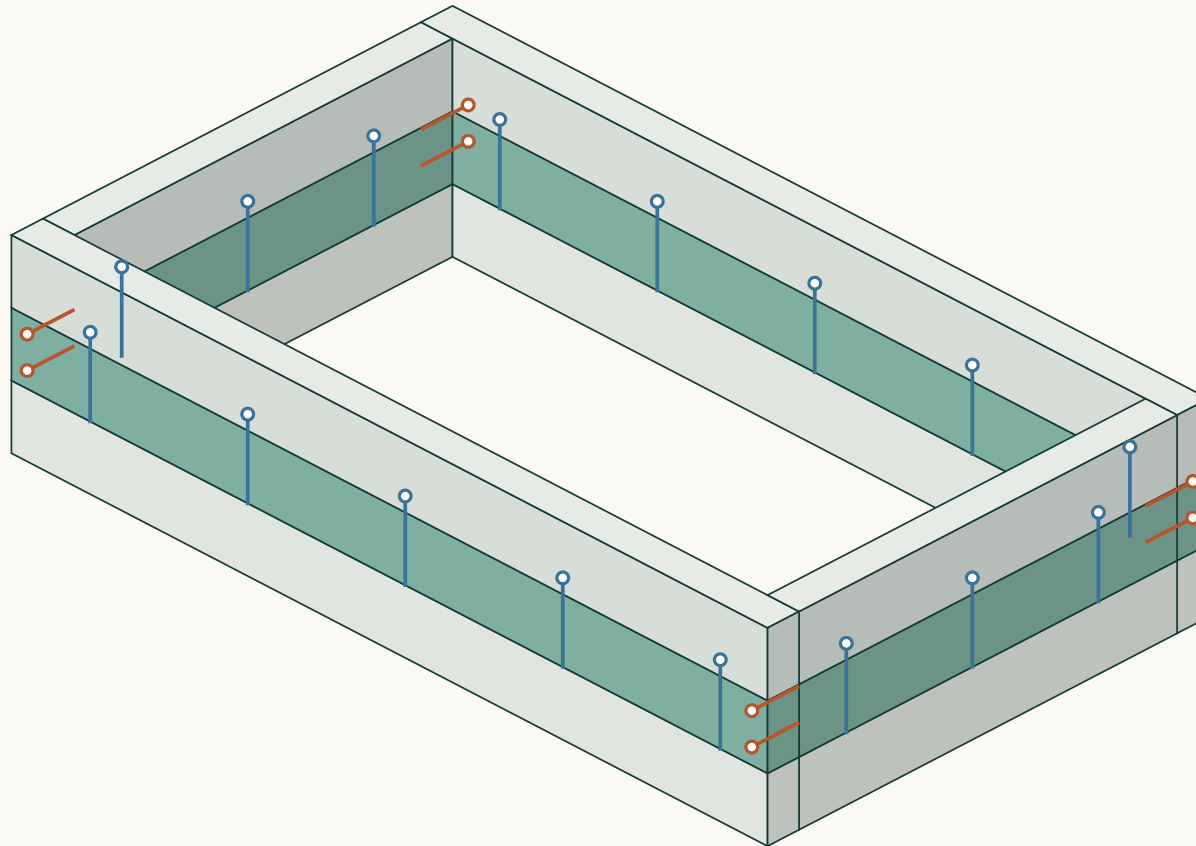
HEAD-SEAT HARD STOP

Course-2 stack screw heads sit on a surface that course 3 must bear on. The current software does not model head diameter, head height or recess. Confirm that detail before covering these heads.

11 / COMPLETE COURSE 2

Eight corner + sixteen stack screws.

Finish the corner fixings, then the vertical schedule. Recheck the layer before course 3 hides the interfaces.



8 CORNER

16 STACK

INSPECT ALL FOUR SIDES

Look for gaps between courses, proud heads, splitting, wandering screws and movement at the corners. A clean visual inspection belongs in the repeatable process.

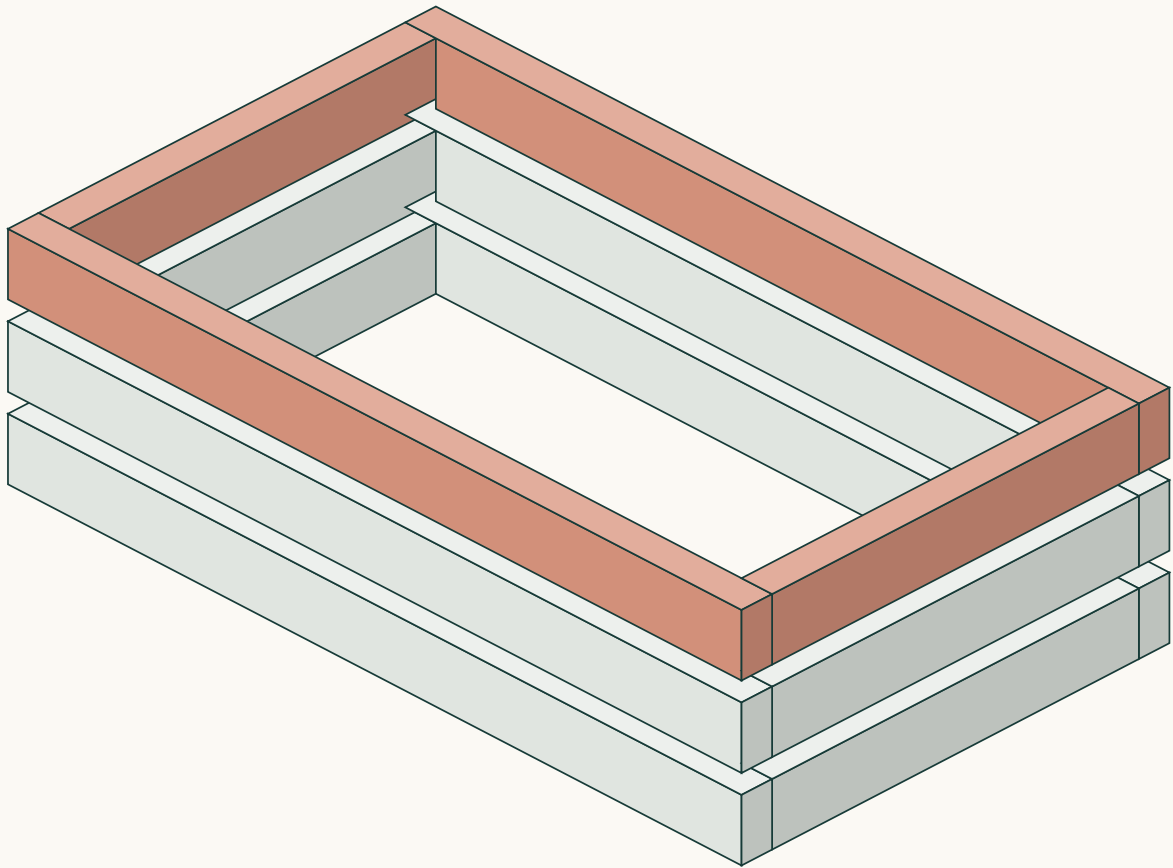
ONLY THEN ADD COURSE 3

The next layer is deliberately staggered relative to the vertical screw rows below. Do not simply copy the course-2 template.

12 / PLACE COURSE 3

Final ring, same corner layout.

Course 3 uses the last two 2400 members and last two 1200 cross-members.



PART	LENGTH	BOARD
C3-S	2400	B006
C3-N	2400	B005
C3-W	1200	B012
C3-E	1200	B011

DO NOT COPY COURSE 2

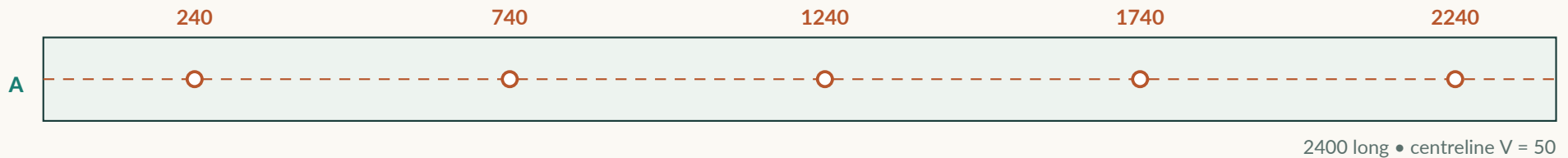
The course-3 vertical entries move 40 mm along each piece in the current model. That stagger avoids the nominal shafts below.

13 / MARK THE STAGGERED TOP ROW

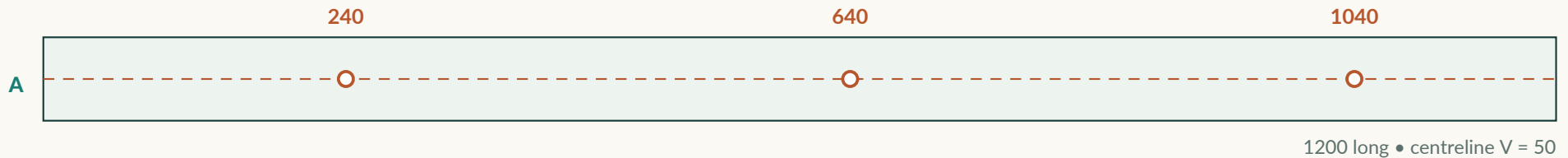
Course 3 shifts by 40 mm.

The same $V = 50$ centreline applies, but every U value moves $+40$ relative to course 2.

S + N / EACH 2400 MEMBER



W + E / EACH 1200 MEMBER



WHY THE SHIFT EXISTS

The software generates all corner paths first, then positions stack fixings to avoid modelled shafts. Course 3 therefore uses a different mark-out row from course 2.

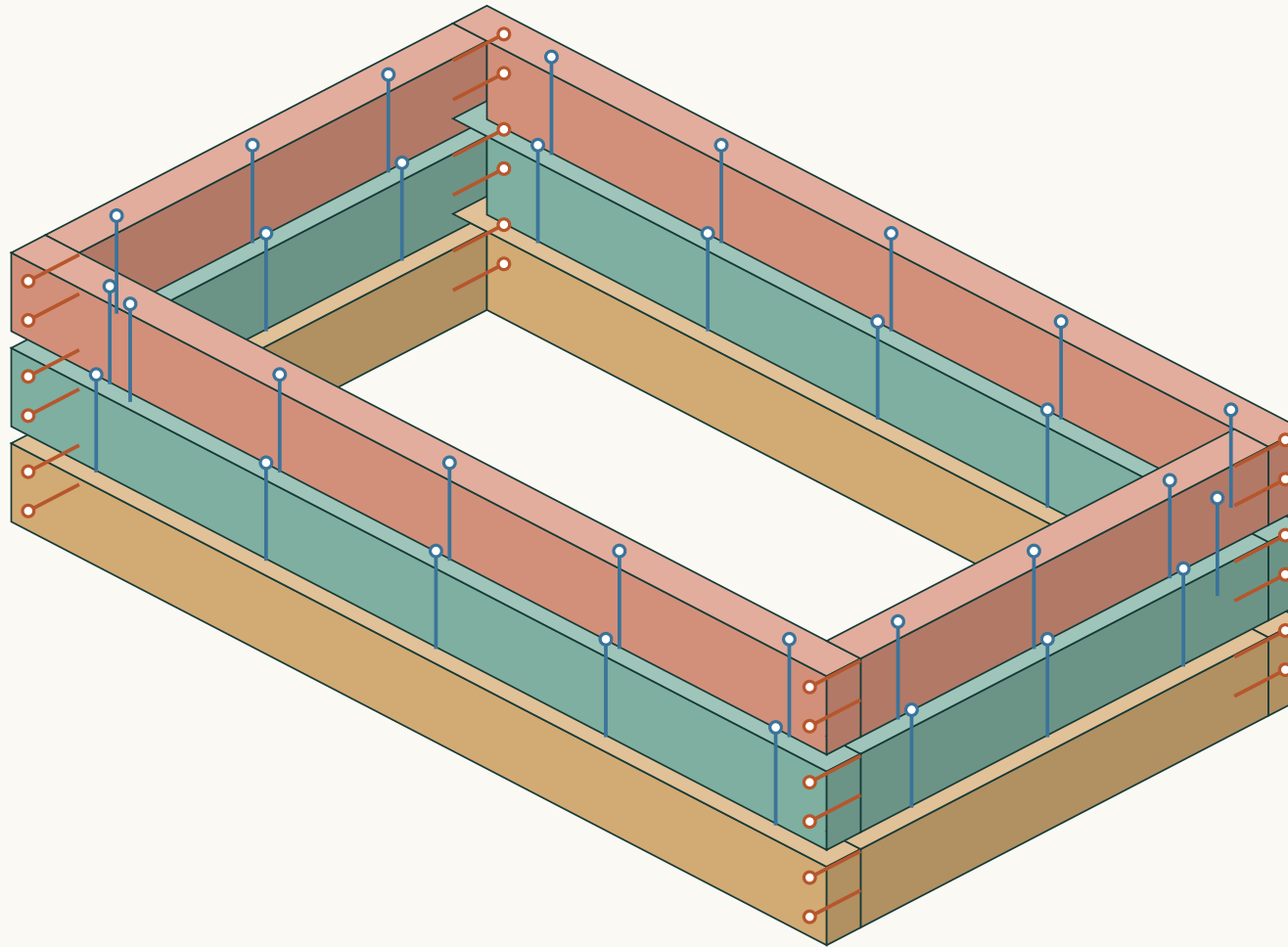
DO NOT GENERALISE THE 40 mm

That offset belongs to this exact geometry and screw-path model. A different footprint, timber section, screw length or recess detail can require a different schedule.

14 / SEE ALL 56 FIXINGS

The screw map is easier to trust when you can see the whole system.

Orange = corner screws through the visible outer faces. Blue = vertical stack screws from courses 2 and 3.



24 CORNER

32 STACK

COUNT BEFORE YOU FINISH

Course 1: 8. Course 2: 24. Course 3: 24. Total = 56 fitted screws. The purchase allowance is larger than the fitted count.

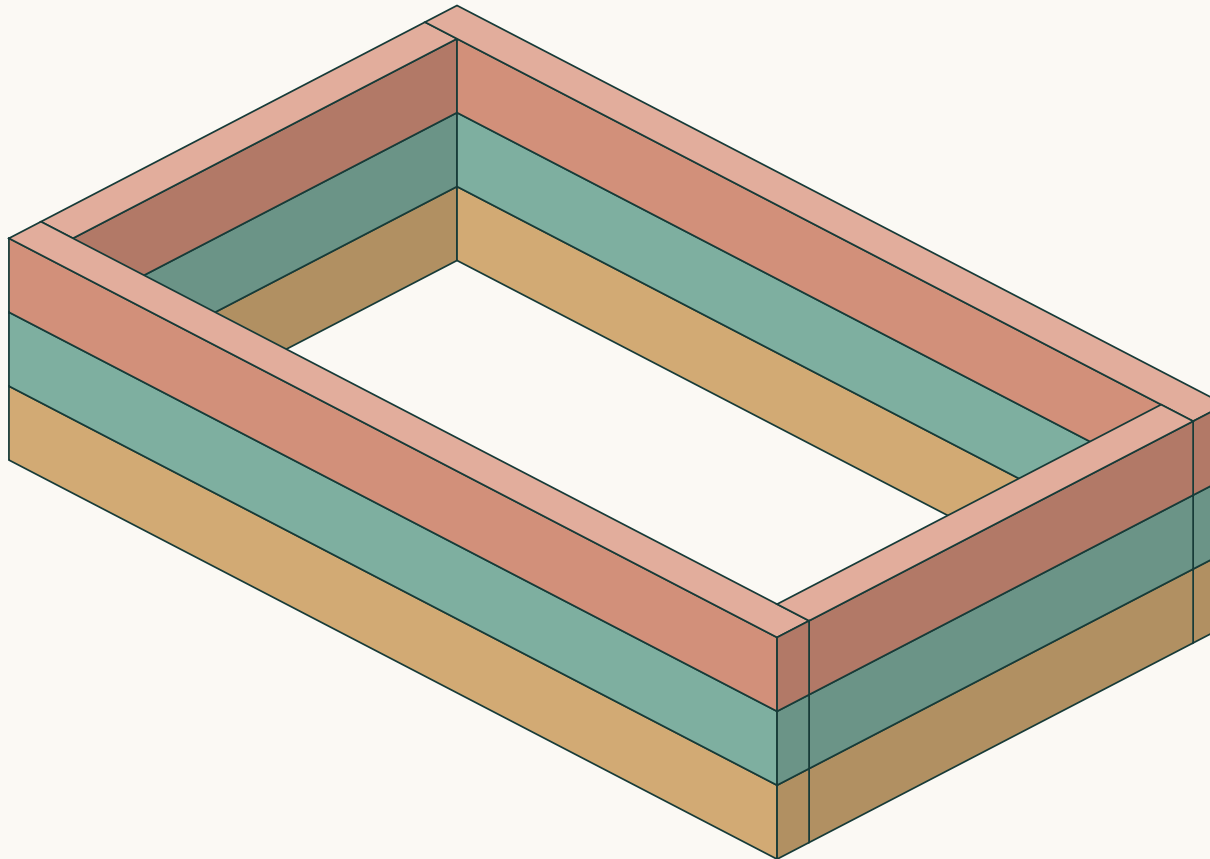
NOT A STRENGTH CHECK

Placement only: no soil-load, screw-capacity, bracing or foundation calculation.

15 / FINISHED TIMBER SHELL

The assembled bed should read as three clean courses.

Before liner, fill or planting, inspect the geometry and all visible interfaces while corrections are still possible.



FINAL CHECK

- ✓ Outside 2400 × 1400
- ✓ Top height 600
- ✓ Both diagonals 2778.5
- ✓ All 12 parts labelled / reconciled
- ✓ 24 corner screws accounted for
- ✓ 32 stack screws accounted for
- ✓ No unresolved proud-head interference
- ✓ Cut ends treated per timber requirement
- ✓ Separate support/drainage detail complete

16 / THE WHOLE BUILD IN ONE PAGE

A compact workshop sequence.

Use this page as the pre-flight card; use the detailed pages for actual mark-out.

1



Measure / verify timber and site

2



Label B001-B012 and every A datum

3



Cut B007-B012 at the 1200-1203 waste band

4



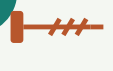
Dry-fit course 1 square and level

5



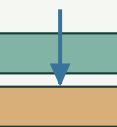
Mark / prepare corner centres from the approved joint recipe

6



Drive 8 course-1 corner screws

7



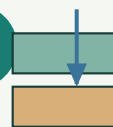
Place course 2, mark 16 stack entries + 8 corners

8



Prepare / drive course 2; inspect head seating

9



Place course 3, use the +40 staggered top schedule

10



Drive course 3; reconcile all 56 fitted screws

11



Final geometry + interface inspection

12



Only then proceed to reviewed supports / drainage / fill

STOP IF THE PACK STILL SAYS TBC

Do not drill until pilot and head-seat operations are verified for the actual screw, timber and joint.

17 / EXACT MARK-OUT SUMMARY

The numbers behind the pictures.

All centres below are local coordinates from the labelled A end of the piece.

COURSE / PIECE	CORNER U / W	STACK U	V
C1-S, C1-N	U 50, 2350 × W 50, 150	-	outer face
C1-W, C1-E	receivers only	-	-
C2-S, C2-N	same corner pattern	200, 700, 1200, 1700, 2200	50
C2-W, C2-E	receivers only	200, 600, 1000	50
C3-S, C3-N	same corner pattern	240, 740, 1240, 1740, 2240	50
C3-W, C3-E	receivers only	240, 640, 1040	50

CORNER W COORDINATE

The two corner centres are 50 and 150 mm up the 200 mm outer face. The long member carries four entries total: two heights at each end.

STACK W COORDINATE

All stack entries begin on TOP: local W = 200. Their screw direction is vertically down. “200 through + 50 penetration” is not a pilot depth instruction.

Nominal screw path: corner = 7 × 150, 100 through + 50 into receiver. Stack = 7 × 250, 200 through + 50 into receiver. Pilot diameter, pilot depth and head-seat dimensions intentionally remain TBC.

18 / WHAT IS STILL TBC

The pack is visually complete; the joint recipe is not.

These are the remaining pieces of information needed before this becomes a drill-ready issued construction manual.

PILOT / CLEARANCE

Exact bit diameter(s), depth(s), which member each operation prepares, and the depth datum.

HEAD SEAT

Whether the washer head remains proud, beds flush by design, or uses a specifically permitted recess; dimensions and driver access.

CONNECTION REVIEW

Suitability of the selected screw for treated timber, end-grain corner joint and the required connection capacity.

SUPPORT / DRAINAGE

Base preparation, restraint/bracing, liner/drainage details and actual site conditions.

PHYSICAL TRIAL

One controlled corner and one stacked interface built with the real timber and hardware; record fit, finish and any corrections.

Traceability: source commit 37db13b378fc868f2d4e6b73f8ba19f6e5f4b526. Core geometry transcribed from the 2400 × 1400 long-through model in the repo. Exact parts/cuts/fixings checked against the committed r08 build artifacts. Review-model SHA-256: 94ab5b0d78e8be414a28ee019e70a8edafd86a35f47ce2b3b14fe2b066fcbe2d.